

AIC paper SOP

1. Frame the paper

What to do

Define the paper as a response to a **reader's problem of understanding**, not just a topic. Show an **instability** the reader should care about, then show why resolving it matters in practice.

Check

- Who is the AIC reader?
- What do they want to understand better?
- What is unstable, incomplete, unreliable, hard to compare, or weakly validated?
- Why does that matter for construction, inspection, monitoring, control, productivity, safety, or lifecycle outcomes?
- Is this a **gap** problem or an **error** problem?

2. Structure the paper

What to do

Use the AIC sequence: **Abstract** → **Introduction** → **Related work** → **Materials and methods** → **Results** → **Discussion** → **Conclusions**. Keep **robustness / uncertainty** and **limitations / quality of evidence** visible.

Check

- Does every section have one clear job?
 - Is there a visible place for robustness, failure cases, limitations, evidence quality, and confidence?
 - Does the structure lead to a bounded conclusion, not just a headline claim?
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Section SOP

3. Abstract

What to do

State: context, current limitation, why it matters, method/system, data or scenario, main finding, bounded conclusion.

Check

- Does it state a real instability?
- Does it state practical relevance?

- Does it state only findings actually shown later?
- Is the conclusion bounded to the tested setting?

4. Introduction

What to do

Do this in order: context, current instability/problem, why it matters, objective, numbered contributions, paper organization. The introduction should construct the problem, not rely on background alone.

Check

- Is the problem explicit and practical?
- Is the objective a direct response to the stated problem?
- Does each contribution have evidence later in the paper?
- Have you avoided overclaiming before showing evidence?

5. Related work

What to do

Organize by **problem tension**, not by catalogue. Show what prior work achieved, where it becomes unstable or hard to compare, and what exact unresolved problem remains.

Check

- Does each subsection revolve around a tension rather than a technology list?
- If you say the literature is inconsistent, weakly validated, or hard to compare, have you shown why?
- Is the gap genuine, or only your framing?

6. Materials and methods

What to do

Include: problem definition, data/case setting, method/system, baselines, experimental design, metrics, sensitivity/robustness plan, reproducibility details.

Add a **statistical design step** for the main claims:

- define the **main comparison**
- define the **main outcome(s)**
- define the **main effect size(s)**
- state which **statistical tests** will be used
- report **confidence intervals**
- report **p-values**
- explain what would count as a **practically meaningful** effect, not just a statistically detectable one

Add a **causal design step** only when the paper truly uses causal language. First decide whether the study is asking an **association**, **intervention**, or **counterfactual** question.

Check

- Is the unit of analysis clear?
- Is the downstream use case clear?
- Are data source, sampling, labels/ground truth, and realism clear?
- Are controllable variables, defaults, ranges, and assumptions reported?
- Are comparators fair and sufficiently controlled?
- What exact claim is each experiment meant to test?

Claim-defense check

- What sensitivity tests will challenge the claim?
- What limitations of model, dataset, assumptions, and validation setting already constrain the claim?
- How will evidence quality be judged?
- What level of confidence would count as justified if the results hold?

Statistical check

- What is the main estimate?
- What is the main effect size?
- What CI will be reported?
- What null or comparison is the p-value testing?
- Are you planning to report p-value, CI, and effect size together rather than using significance alone?
- What result would be statistically detectable but still too small to matter in practice?

Causal check

- Is this study only descriptive, or is it estimating an intervention effect?
- If causal language is used, what is the causal question?
- What assumptions make the causal claim identifiable?
- If a counterfactual claim is made, is the design strong enough for it?

6A. Polanyi paradox check

What to do

Check whether the target task depends on expert judgement that is used reliably but not fully stated as rules. If yes, state clearly:

- what expert knowledge is made explicit as rules, priors, features, constraints, or workflow steps
- what is only approximated through data, examples, or learned models
- what still remains tacit and requires review, abstention, or deployment limits

Bound the paper's claim accordingly: the method may reduce reliance on expert judgement without fully replacing it.

Check

- Does the task rely on tacit expertise?
- What has been formalized directly?
- What has only been approximated?
- What still requires human judgement?
- Did the method remove expert dependence, reduce it, or relocate it?
- Are the paper's claims bounded to the part of expertise actually captured?
- Have you avoided claiming full automation if tacit judgement still remains?

7. Results

What to do

Report: main quantitative results, component/ablation analysis, sensitivity/robustness results, error/failure cases, and practical performance.

For the main comparisons, report results in this order:

1. the **estimate / contrast**
2. the **effect size**
3. the **confidence interval**
4. the **p-value**
5. the **practical meaning**
6. the **sensitivity result**

If causal claims are being made, separate:

- **associational results**
- **interventional effect estimates**
- **counterfactual analyses**, if justified

Check

- Is the main comparison clear?
- Are uncertainty/spread shown, not just averages?
- Does each claimed component contribution have a matching controlled comparison?
- Are relevant sensitivity results visible?
- Are failure cases shown and explained?
- Are runtime, memory, scalability, or deployment constraints reported where relevant?

Claim-defense check

For each major result:

- What sensitivity evidence supports it?
- What limitation weakens it?
- How strong is the underlying evidence?
- What confidence level is justified: high, moderate, or low?

Statistical check

- Have you shown the estimate before the p-value?

- Is the CI shown with the estimate?
- Is the effect size interpretable on the application scale?
- Are exact p-values reported?
- Are statistically significant but practically trivial findings identified as such?
- Are practically important but statistically uncertain findings discussed honestly?

Causal check

- Which results are descriptive only?
- Which are intervention claims?
- If counterfactual results are shown, are they clearly labeled as inferred rather than observed?
- Are design diagnostics visible enough for the reader to trust the causal estimate?

8. Discussion

What to do

Interpret the findings, compare them with prior work, explain implications for AIC readers, state limitations directly, and identify future work that follows only from demonstrated limitations.

Discuss results in this order:

- what the estimate suggests
- how large the effect is
- how uncertain it is
- whether it is practically important
- how robust it is
- what limitations weaken it
- what confidence the reader should place in it

Check

- Are you interpreting, not repeating?
- Are comparisons with prior work fair?
- Are practical implications actually supported by the evidence?
- Have you named the limits of model, data, assumptions, and validation setting?

Claim-defense check

For each main conclusion, summarize:

- sensitivity evidence
- main limitations
- evidence quality
- confidence level
- where caution is needed

Statistical check

- Are you discussing the estimate, not just the p-value?
- Are you discussing the width and location of the CI?
- Have you separated statistical significance from practical significance?
- Have you said clearly when the evidence is precise but the effect is small?
- Have you said clearly when the effect may matter but uncertainty remains large?

Causal check

- Has the discussion stayed at the right rung: association, intervention, or counterfactual?
- Are assumptions treated as part of the result, not a footnote?
- Is mechanism discussed only if actually tested?
- Are transportability and external-validity limits stated carefully?

9. Conclusions

What to do

Use the AIC pattern: one sentence on the problem, one on the method, one on the main evidence, one on practical meaning, one bounded takeaway.

Check

- Does the conclusion match the evidence rather than the ambition?
- Is the final takeaway bounded?
- Have confidence and caution been preserved?

Global paper checks

10. Claim-defense checklist

For every major claim, ask:

- What is the claim exactly?
- What direct evidence supports it?
- What sensitivity tests challenge it?
- What model/data/assumption/validation limits constrain it?
- How strong is the evidence quality?
- What confidence level is justified?
- Is the wording matched to that confidence?

11. Statistical-analysis checklist

For every main quantitative claim, ask:

- What is the **estimate**?
- What is the **effect size**?

- What is the **confidence interval**?
- What is the **p-value**?
- What is the **practical significance**?
- Does the result remain under sensitivity analysis?
- Is the wording driven by the estimate and uncertainty, rather than by significance alone?

12. Causal-language checklist

If you use words like **effect, impact, leads to, drives, explains, mechanism, would have happened**, check the causal rung first:

- **Association**: observed pattern
- **Intervention**: what happens if X is changed
- **Counterfactual**: what would have happened otherwise

13. Wording checklist

- Use **shows** only when strongly supported.
- Use **indicates** or **suggests** when evidence is narrower.
- Use **in the tested setting** when generalization is limited.
- Do not claim robustness, generalizability, practicality, usability, or causality without direct support.
- Do not use **$p < 0.05$** as the main defense of a claim.
- Do not say **no effect** just because the result is not statistically significant; use the CI and effect size to explain what remains uncertain.

Daily-use final checklist

Before finalizing any section, ask:

1. Who is the AIC reader?
2. What exact instability does this section address?
3. Why does it matter in practice?
4. What claim is being made?
5. What evidence supports it?
6. What sensitivity tests challenge it?
7. What limitations constrain it?
8. How strong is the quality of the evidence?
9. What confidence is justified?
10. What are the estimate, CI, effect size, and p-value for the main quantitative claim?
11. If causal language is used, is it association, intervention, or counterfactual?

Master rule

Build the paper around a real reader's problem, and at every stage defend each claim with robustness, limitations, evidence quality, confidence, and—when needed—the correct level of statistical and causal reasoning.

we do it later. now we need redesign and rerun our experiment to support the paper's thinking flow. we already did a lot of pre-experiment to test the validity of our concept, please revisit all the previous logs and journals and redesign our datasets, experiment, ablation study, evaluation metrics, LLM consistency, sensitivity study...etc. @methods/journals @logs

the design principle is "Claims need sensitivity analysis, recognition of their limitations, and summarise of the quality of the underlying evidence, and preferably assessments of the confidence in the whole analysis."